



ENTERPRISE SYSTEM ARCHITECTURE REPORT

Cinnamon Heritage (Pvt) Ltd

Complete A-to-Z Technical Breakdown, Media Cloud Integration, Firestore Database Schema, Function Directory, and 100% CMS Customizability Guide.

SYSTEM EDITION

Production Cloud v2.0

MEDIA & CDN HOSTING

Cloudinary High-Speed
CDN

CLOUD DATABASE

Google Cloud Firestore

AUTHENTICATION

Firebase Auth (Multi-
Role)

DEVELOPMENT ENGINE

Node.js Runtime &
Vercel Edge

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Cinnamon Heritage is an enterprise-grade digital platform designed to manage the supply chain, retail commerce, export inquiries, and hospitality operations of an authentic Ceylon Cinnamon (*Cinnamomum verum*) estate in Galle, Southern Sri Lanka.

 Direct Retail E-Commerce

Customer storefront with real-time stock deductions, multi-step progress tracking, discount rules, and payment toggles.

 B2B Wholesale Export

Dedicated bulk inquiry engine with automated certificate selection, custom specifications, and automated email dispatches.

 Tourism & Eco-Villa Bookings

Reservation pipeline for artisan processing demonstrations, factory tours, and eco-villa stays.



ARCHITECTURE LAYER	TECHNOLOGY / LIBRARY	VERSION / DELIVERY	KEY RESPONSIBILITIES
UI Presentation	HTML5 + CSS3 Custom	Standard	Glassmorphic luxury dark theme, responsive grid, SEO JSON-LD schema markup.
Component Framework	Bootstrap Bundle	5.3.2 (CDN)	Responsive layout grid, Modals, Offcanvas drawers, Toast alerts, Dropdowns.
Icons & Typography	FontAwesome & Bootstrap Icons	v6.0 / v1.11	UI action buttons, status icons, payment logos, Oswald & Inter typography.
Cloud NoSQL Database	Google Cloud Firestore	Compat v10.8.0	Real-time document collections, ACID order transactions, snapshot streams.
Identity & Access	Firebase Authentication	Compat v10.8.0	Secure token-based auth for Customers and Role-Gated Admin Accounts.
High-Res Media CDN	Cloudinary REST API	Direct Streaming	Direct client-to-cloud asset hosting with real-time percentage progress bars.
Business Analytics	Chart.js	v4.x (CDN)	Interactive revenue trends and payment method distribution charts.
Email Gateway	EmailJS Browser SDK	v4.x	Automated dispatches for B2B export leads and Experience tour inquiries.

ARCHITECTURE LAYER	TECHNOLOGY / LIBRARY	VERSION / DELIVERY	KEY RESPONSIBILITIES
Dev Runtime & Hosting	Node.js & Vercel Edge	Node 18+ / Vercel Static	Local development server (`npx serve`) and edge CDN production hosting.



To ensure high performance and eliminate server disk consumption, all image uploads across the platform (product images, section full-width backgrounds, step illustrations, and user avatars) are offloaded to Cloudinary's global media network.

Direct-to-Cloud Upload Streaming

Admin uploads do not pass through intermediate web servers. Uploads stream directly via an unsigned upload preset to Cloudinary using `XMLHttpRequest`, which provides continuous byte-level progress updates (0% to 100%).

Edge Optimization & Security

Images are automatically delivered in modern WebP format with responsive dimensions, saving over 70% bandwidth while maintaining high resolution on Retina and 4K displays.

```
// Cloudinary Upload Engine Signature (js/firebase.js)
function uploadImageWithProgress(file, onProgress) {
  // Dispatches multipart/form-data to Cloudinary REST endpoint
  // Fires onProgress(Math.round((loaded / total) * 100)) during transmission
  // Resolves with high-speed HTTPS CDN URL: https://res.cloudinary.com/...
}
```



The system organizes business entities into 9 primary NoSQL collections in Google Cloud Firestore:

COLLECTION NAME	PRIMARY KEY	FIELD SCHEMA OVERVIEW	REAL-TIME LISTENERS ACTIVE
profiles	User Auth UID	email , role ('admin' 'customer'), is_approved (bool), status ('pending' 'approved'), cart_items (array), mobile , city , created_at	SESSION AUTH
products	Auto ID	title , price , discount , stock_quantity , total_sales , delivery_charge , image_url , is_active , display_order	ONSNAPSHOT
orders	#CH- Timestamp- Rand	customer_name , customer_email , shipping_address , district , payment_method , total_amount , status , order_items , created_at	ONSNAPSHOT (USER & ADMIN)
site_settings	Key Name	cod_enabled , bank_transfer_enabled , online_pay_enabled , bank_name , bank_branch , bank_acc_no , low_stock_threshold	ONSNAPSHOT
site_content	Section Name	hero , about , villa , factory , process , products , b2b ,	ONSNAPSHOT

COLLECTION NAME	PRIMARY KEY	FIELD SCHEMA OVERVIEW	REAL-TIME LISTENERS ACTIVE
		essential_oils , experience , faq (Each contains title, text, background_image)	
delivery_rates	District	province , district , rate (LKR)	CACHED LOAD
process_steps	Step ID	title , description , image_url , display_order	ONSNAPSHOT
faqs	FAQ ID	question , answer , display_order	ONSNAPSHOT
b2b_enquiries	Auto ID	company_name , contact_person , email , phone , volume , certificates , status	ADMIN QUERY



Every single visual and textual component on the storefront is fully customizable from the Admin Panel without code changes. Background images can be uploaded directly or linked via URL:

STOREFRONT SECTION	EDITABLE TEXT & FUNCTIONAL FIELDS	BACKGROUND IMAGE CMS CONTROLS	MEDIA UPLOAD GATEWAY
1. Hero Banner	Headline, Subtitle, CTA Button Text & Destination URL, Video Preview Modal URL	Full-width Hero Background Image	CLOUDINARY + PROGRESS BAR
2. About Estate	Founding Story, Heritage Timeline, Estate Acreage, Mission Statement	Full-width About Us Background Image	CLOUDINARY + PROGRESS BAR
3. Cinnamon Villa	Eco-Villa Description, Tour Packages, Amenities List, Reservation Link	Full-width Villa Background Image	CLOUDINARY + PROGRESS BAR
4. Factory & Lab	Processing Standards, Steam Distillation Specs, Secondary Feature Image	Factory Background Image + Feature Image	CLOUDINARY + PROGRESS BAR
5. 8-Step Process	Step 1 to 8: Step Title, Detailed Explanation, Display Ordering	Process Section Background + 8 Step Images	CLOUDINARY + PROGRESS BAR
6. Products Catalog	Product Name, Price, Discount %, Stock Quantity, Active Toggle, Card Design	Shop Catalog Section Background Image	CLOUDINARY + PROGRESS BAR
7. B2B Wholesale	Partner Categories, Packaging Options, Export Certificates, Notification Email	Full-width B2B Background Image	CLOUDINARY + PROGRESS BAR

STOREFRONT SECTION	EDITABLE TEXT & FUNCTIONAL FIELDS	BACKGROUND IMAGE CMS CONTROLS	MEDIA UPLOAD GATEWAY
8. Essential Oils	Leaf vs Bark Oil Specs, Eugenol/Cinnamaldehyde %, Inquiry Destination	Essential Oils Section Background Image	CLOUDINARY + PROGRESS BAR
9. Tours & Experience	Artisan Peeling Workshops, Estate Walkthroughs, Pricing, Booking Email	Experience Section Background Image	CLOUDINARY + PROGRESS BAR
10. Interactive FAQs	Q&A Content, Display Sequence, Add/Edit/Delete Accordion Items	Full-width FAQ Section Background Image	CLOUDINARY + PROGRESS BAR
11. Footer & Socials	Hotline, WhatsApp, Email, Office Address, Facebook, Instagram, LinkedIn	Managed dynamically via Dynamic Footer CMS	DIRECT FIRESTORE SETTINGS
12. Payments & Freight	COD (ON/OFF), Bank Transfer (ON/OFF + Bank Credentials), Pay Now (ON/OFF)	District-by-District Delivery Pricing Rules	FIRESTORE SITE SETTINGS



index.html — Public Storefront

- Preloader with 1s maximum safety dismissal and 0.3s fade.
- Hero slider with Cloudinary background & video modal.
- Live product grid with low-stock badges.
- Dual-tab Cart Drawer (Cart Items & Live Order Tracking Stepper).
- Customer/Admin Authentication Modal with password visibility toggle.

checkout.html — Checkout Engine

- Automated cart ('?cart=true') vs single item buy detection.
- Live payment gateway switches (COD / Bank Transfer / Pay Now).
- Dynamic bank transfer details presentation with reference slip upload.
- District freight recalculation on dropdown change.
- ACID inventory decrement & auto cart clearing upon order completion.

admin.html — Admin Enterprise Portal

- 5 Equalized Top KPI Cards (Earnings Today, Month, Orders, In Progress, Products).
- Quick Stats with This Year's & Lifetime Earnings.
- Order Management with 7 summary filter cards & "Update Process" modal.
- CMS Editors for all 10 sections and Cloudinary background image uploads.
- Admin Account Governance with pending approval gate.



Every core JavaScript module is organized cleanly with single-responsibility functions:

 A. Cloud & Data Layer — `js/firebase.js`

FUNCTION SIGNATURE	PARAMETERS	DESCRIPTION & IMPLEMENTATION DETAILS
<code>uploadImageWithProgress(file, onProgress)</code>	<code>file</code> (File), <code>onProgress</code> (callback)	Streams image to Cloudinary REST endpoint with 0-100% upload progress tracking.
<code>uploadImage(file)</code>	<code>file</code> (File)	Wrapper for image uploads returning the resolved Cloudinary HTTPS URL.
<code>placeOrder(orderData)</code>	<code>orderData</code> (Object)	Runs a Firestore ACID Transaction (<code>db.runTransaction</code>) to verify stock, atomically decrement inventory, generate order ID, and record transaction.
<code>subscribeToUserOrders(email, callback)</code>	<code>email</code> (string), <code>callback</code> (function)	Attaches a Firestore <code>onSnapshot</code> listener to provide real-time order status updates to the customer's cart drawer without page refresh.
<code>getDashboardStats()</code>	None	Aggregates financial KPIs: <code>earningsToday</code> , <code>earningsMonth</code> , <code>earningsYear</code> , <code>totalEarnings</code> , and <code>inProgressOrders</code> .

FUNCTION SIGNATURE	PARAMETERS	DESCRIPTION & IMPLEMENTATION DETAILS
<code>serveFromCache(key, fetcher)</code>	<code>key</code> (string), <code>fetcher</code> (function)	Serves cached JSON from <code>sessionStorage</code> instantly while fetching fresh data in the background.
<code>authLogin(email, password)</code>	<code>email</code> , <code>password</code>	Authenticates user via Firebase Auth and checks approval status if user is an admin.
<code>authSignup(username, email, password, extra)</code>	<code>username</code> , <code>email</code> , <code>password</code> , <code>extraData</code>	Creates auth record and stores user profile in Firestore. Sets <code>is_approved: false</code> for admin registrations.
<code>saveContentFields(section, fields)</code>	<code>section</code> (string), <code>fields</code> (Object)	Saves section text, settings, and Cloudinary background image URLs to Firestore.
<code>saveSetting(key, value)</code>	<code>key</code> (string), <code>value</code> (any)	Updates global e-commerce configs (e.g. COD switch, Bank transfer details).

 B. Client UI & Interactivity — `js/main.js`

FUNCTION SIGNATURE	DESCRIPTION & UI INTERACTION
<code>hidePreloader()</code>	Enforces a strict 1-second maximum preloader timeout with a 0.3s snappy CSS fade-out.
<code>setupUserOrdersTracking(email)</code>	Initializes real-time Firestore order subscription for logged-in or guest users.
<code>renderUserOrders(orders)</code>	Renders Amazon/Daraz style 5-step animated progress stepper cards inside the Cart Drawer.

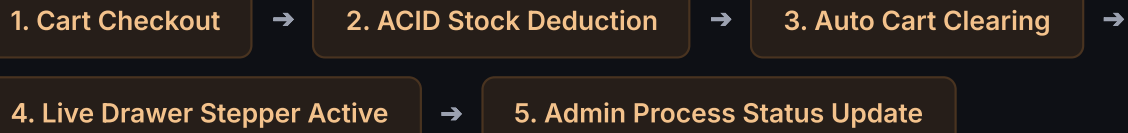
FUNCTION SIGNATURE	DESCRIPTION & UI INTERACTION
<code>updateCartUI()</code>	Synchronizes cart badges, item count, line discounts, and subtotal across desktop and mobile headers.
<code>addToCart(productId)</code> / <code>changeCartQty(id, delta)</code>	Updates cart in <code>localStorage</code> (guest) or Firestore <code>profiles</code> (authenticated).
<code>applyHeroBackground(content)</code>	Applies dynamic Cloudinary hero background image with smooth CSS transition.

 C. Admin CMS & Orders Engine — `js/admin.js`

FUNCTION SIGNATURE	DESCRIPTION & ADMINISTRATIVE ROLE
<code>loadDashboard()</code>	Renders 5 equal KPI cards, This Year's earnings, inventory health alerts, and Chart.js graphs.
<code>loadOrders()</code> & <code>renderFilteredOrders()</code>	Renders order list with multi-criteria filtering (Today, This Month, This Year, Custom Date, Status, Search).
<code>viewOrder(id)</code>	Opens order modal with the Update Process action to transition order states.
<code>updateOrderStatusAndRefresh(id, status)</code>	Updates status in Firestore (<code>Pending</code> , <code>In Progress</code> , <code>Packed</code> , <code>Shipped</code> , <code>Completed</code> , <code>Cancelled</code>).
<code>handleProductImageUpload(id)</code>	Uploads product photos to Cloudinary with live progress bar feedback in Admin table.
<code>loadAdminAccounts()</code> & <code>approveAccount(uid)</code>	Admin security approval gate to activate or revoke admin accounts.



A. Real-Time Order & Tracking Lifecycle



Customer Live Stepper (Daraz/Amazon Style)

When an order is placed, customer cart automatically clears and redirects to the "My Orders" tab inside the drawer. The 5-stage stepper (`Pending` → `In Progress` → `Packed` → `Shipped` → `Completed`) pulses in real time as the admin updates the status.

Admin Account Security Gate

Self-registered admins cannot access the portal immediately. New accounts receive `is_approved: false` and `status: 'pending'`. An existing Super Admin must approve them from the Admin Accounts panel before login is granted.



ACID Inventory Integrity

All orders run inside Firestore transactions to prevent race conditions, double-orders, or negative inventory states.

Sub-Second Preloader

1-second maximum hard limit with session caching guarantees snappy user experience on high-latency mobile networks.

Zero Server Bloat

All media assets are securely hosted on Cloudinary CDN, ensuring zero server disk consumption and optimal bandwidth usage.

Cinnamon Heritage (Pvt) Ltd — System Architecture Documentation & Technical Report

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